

HASTINGS, NE, USA

WMO: 725525

Lat: 40.601N

Lon: 98.426W

Elev: 598

StdP: 94.35

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94949

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-16.2	-23.0	0.5	-17.9	-20.8	0.6	-15.2	15.5	-2.6	13.6	-1.9	4.5	340	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	35.1	22.8	33.4	22.7	32.1	22.4	25.1	31.7	24.3	30.7	23.4	29.8	5.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	19.1	28.7	22.4	18.4	28.1	21.4	17.3	27.1	80.2	31.8	76.4	30.7	72.8	29.6	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.1	9.5	DB	-23.0	37.9	2.2	1.4	-24.6	38.9	-25.9	39.7	-27.2	40.5	-28.8	41.5	(4)
(5)			WB	-23.3	26.9	2.2	1.1	-24.9	27.7	-26.2	28.3	-27.4	28.9	-29.0	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-3.2	-1.4	4.7	10.5	16.4	22.4	25.0	23.4	19.2	11.7	4.4	-1.9	(6)	
	DBStd	11.26	6.69	6.87	6.46	5.43	4.83	3.73	3.24	3.09	4.52	5.35	5.73	6.01	(7)	
	HDD10.0	1576	410	322	187	58	6	0	0	0	1	42	181	369	(8)	
	HDD18.3	3292	668	553	423	240	96	9	1	2	43	212	417	627	(9)	
	CDD10.0	1944	0	2	23	72	205	372	466	416	277	97	14	0	(10)	
	CDD18.3	617	0	0	1	4	36	131	209	159	69	8	0	0	(11)	
	CDH23.3	6265	0	0	16	82	399	1335	2199	1442	688	98	4	0	(12)	
(13)	CDH26.7	2593	0	0	2	21	133	544	1032	578	255	27	1	0	(13)	
(14)	Wind	WSAvg	4.5	4.6	4.8	5.2	5.6	4.9	4.4	3.7	3.5	4.1	4.5	4.5	(14)	
(15) Precipitation	PrecAvg	724	17	21	48	71	120	104	94	83	79	43	31	19	(15)	
	PrecMax	1221	87	75	208	187	335	260	291	264	339	122	101	94	(16)	
	PrecMin	454	0	1	2	1	16	21	4	12	6	0	0	0	(17)	
	PrecStd	162	15	19	41	40	60	63	62	54	66	30	29	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	19.6	25.5	29.2	32.8	36.1	37.8	36.4	34.5	30.1	23.3	15.3	(19)	
		MCWB	7.3	9.3	13.5	15.7	19.7	22.1	23.2	23.5	20.8	18.7	12.6	7.7	(20)	
	2%	DB	11.5	14.6	22.0	25.9	30.1	33.7	35.8	33.8	32.1	26.0	19.8	11.9	(21)	
		MCWB	5.2	7.6	12.3	15.2	19.3	22.3	23.7	23.2	20.8	15.9	11.0	6.1	(22)	
	5%	DB	8.5	11.3	18.5	22.7	27.8	32.1	33.8	32.1	29.8	23.1	16.4	8.9	(23)	
		MCWB	3.7	6.0	11.0	13.9	18.4	22.0	23.6	23.1	20.2	14.6	9.8	4.4	(24)	
	10%	DB	5.8	7.9	15.4	20.0	25.3	29.9	32.2	29.9	27.4	20.9	13.3	6.3	(25)	
		MCWB	2.3	4.2	9.3	12.5	17.2	21.2	23.2	22.6	19.0	14.0	7.8	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	11.4	16.4	18.0	22.3	25.2	26.4	26.0	23.3	20.1	14.8	9.2	(27)	
		MCDB	14.8	16.3	20.9	24.9	29.0	31.6	33.0	31.6	30.0	26.2	19.0	13.9	(28)	
	2%	WB	5.6	8.4	14.5	16.3	20.9	23.9	25.4	24.8	22.2	18.2	12.4	6.3	(29)	
		MCDB	11.0	14.0	19.3	22.6	27.4	30.6	32.3	30.8	28.9	22.9	17.6	11.1	(30)	
	5%	WB	4.0	6.3	11.9	14.9	19.6	22.8	24.6	23.9	21.2	16.2	10.3	4.6	(31)	
		MCDB	7.9	11.1	17.1	21.6	25.7	29.6	31.4	29.8	27.6	21.5	15.9	8.7	(32)	
	10%	WB	2.4	4.2	9.6	13.3	18.2	21.9	23.8	23.0	20.1	14.6	8.2	2.8	(33)	
		MCDB	5.6	8.0	15.5	19.2	23.6	28.6	30.3	28.8	25.8	19.6	13.4	6.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	10.9	12.6	12.6	11.6	11.6	11.5	11.2	12.5	12.6	12.1	10.3	(35)	
		MCDBR	15.0	15.8	17.9	16.8	15.1	14.0	14.0	13.6	14.7	16.0	16.6	14.4	(36)	
	5% WB	MCWBR	9.8	9.9	10.0	8.8	6.8	5.7	5.6	5.8	6.0	7.9	9.5	9.6	(37)	
		MCDWR	14.0	15.0	15.7	14.8	12.8	12.4	12.2	12.0	12.7	12.8	15.3	13.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.274	0.279	0.306	0.343	0.370	0.383	0.399	0.397	0.361	0.318	0.278	0.269	(40)	
		taud	2.489	2.499	2.453	2.368	2.366	2.376	2.352	2.348	2.403	2.508	2.534	2.504	(41)	
	Ebn at Noon		901	949	951	929	907	891	873	866	877	888	889	879	(42)	
		Edh at Noon	74	86	101	118	121	120	122	119	105	84	71	68	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.23	3.05	4.14	5.22	6.04	6.92	6.76	5.91	4.80	3.45	2.47	1.90	(44)	
	RadStd		0.21	0.23	0.36	0.29	0.37	0.42	0.33	0.33	0.30	0.34	0.17	0.19	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page